Cricket Analytics: Predicting IPL Match Results using Historical Data & Machine Learning

Sai Padma Sriya Pothula
Department of Data Analytics Engineering
George Mason University
Fairfax, VA, USA
spothula@gmu.edu

Sadiya Sayara Chowdhury Puspo
Department of Information Science & Technology
George Mason University
Fairfax, VA, USA
spuspo@gmu.edu

Mohan Adhi
Department of Data Analytics Engineering
George Mason University
Fairfax, VA, USA
amohan20@gmu.edu

I. Project Description and Motivation

Cricket is a complex and changing sport influenced by many things both inside and outside the field. Factors include how well individual players perform, team strategies, conditions of the pitch and weather, and external factors like crowd support and having a home-ground advantage. Unlike some sports where outcomes can be predicted based on fixed parameters, cricket is uniquely challenging because it is unpredictable. In competitions like the Indian Premier League (IPL), where matches are highly competitive and teams have top players from around the world, predicting results becomes even more difficult. The quick nature of the Twenty20 (T20) format adds another layer of complexity. In a short game of just 20 overs, a small mistake can completely change the outcome. Each team has to balance aggressive play with careful strategy to improve their chances of winning. The unpredictability of IPL matches makes them perfect for predictive analytics. This project aims to deal with the challenge of predicting IPL match outcomes by developing a model based on data and machine learning. The goal is to better understand the factors that affect match results by looking at past data. Knowing these factors can help teams and analysts make smarter decisions, like choosing to bat or bowl first after winning the toss, or how to place important players during different parts of the match. Besides predicting match outcomes, this knowledge can also help in assessing player performance, picking teams, and match preparation, leading to more strategic decisions in future IPL tournaments.

II. Research Questions and Hypothesis

To guide the project, the following research questions have been developed. Each question focuses on a specific aspect of cricket match dynamics, and corresponding hypotheses have been formulated to explore the relationships between these variables and match outcomes.

1. How does winning the toss affect the probability of winning a match in the IPL?

Hypothesis: Winning the toss increases a team's likelihood of winning by providing the advantage of making decisions that align with current pitch and weather conditions, thereby giving them a strategic edge.
Rationale: In cricket, the toss outcome allows teams to adjust their strategies based on external factors like weather and pitch conditions. For instance, they may choose to bat first on a deteriorating pitch or bowl first under humid conditions to exploit swing. In the IPL, venues and conditions vary, so winning the toss can provide a significant tactical advantage.

2. What role does performance in the powerplay phase play in determining match outcomes in T20 cricket?

Hypothesis: Teams that achieve higher runs and lose fewer wickets during the powerplay phase are more likely to win, as a strong start provides early momentum and pressure on the opposition.
Rationale: The powerplay in T20 cricket is a crucial period where only two fielders are allowed outside the 30-yard circle, encouraging aggressive batting. A high score during the powerplay sets a solid foundation for the innings, while

early wickets can put the batting team on the back foot. Examining the correlation between powerplay performance and match outcomes can yield insights into how initial momentum affects the entire game.

3. How do individual performance metrics (e.g., strike rate for batsmen and economy rate for bowlers) influence the likelihood of a player being awarded the player of the match?

Hypothesis: Players with superior performance metrics, such as higher strike rates or lower economy rates, have an increased chance of receiving the player of the match award due to their impactful contributions.
Rationale: In T20 cricket, the player of the match award often goes to players who deliver outstanding individual performances. Metrics like strike rate (for batsmen) and economy rate (for bowlers) reflect a player's effectiveness. Understanding which specific metrics correlate most strongly with player awards can highlight what aspects of performance are most valued in T20 cricket.

III. LITERATURE REVIEW

The rise of machine learning in sports analytics has quickly increased, especially in cricket due to the game's complex and unpredictable nature, making accurate predictions both difficult and valuable. The IPL, a well-known T20 cricket league globally, offers a valuable dataset for predictive analysis. Multiple research projects have investigated the use of past data in forecasting game results, player showings, and important game moments [2].

In 2018, Rabindra Lamsal and colleagues utilized Random Forests and Multiple Linear Regression models to forecast IPL game results with a precision of 68.33%. Their strategy concentrated on fundamental match characteristics like toss outcomes, venue of the match, and makeup of the team. Yet, their model's restricted amount of features led to less-than-ideal accuracy, emphasizing the importance of a more extensive dataset incorporating in-game elements like player performance, powerplay dynamics, and death-over performance [3].

Abhishek Naik et al. (random) made another important impact with the development of a dynamic prediction model for One-Day International (ODI) cricket. This model took into account player statistics, bowler performance, and batting order in both pre-match and in-game situations. They employed a mixture of Random Forest, Support Vector Machines (SVM), and additional classification models, offering a more comprehensive perspective of the game. Yet, more validation is needed for these models when applied to T20 formats, especially in the IPL, as the game is faster paced and unpredictable [4].

Singhvi and colleagues (2015) showcased how 16 features and multiple machine learning algorithms like Naive Bayes, SVM, and Random Forest were utilized to forecast match results in various cricket formats. When trained on a dataset of 5,390 T20 matches, their research obtained a 63.89% accuracy rate. Though their study showed the possibility of machine learning in cricket, it also highlighted the necessity for more advanced features, to increase accuracy [5].

Moreover, Swetha and Saravanan (2017) concentrated on recognizing the key factors that impact match results, including pitch conditions, team strength, home-ground advantage, and toss results. They didn't prepare machine learning models but conducted a detailed statistical analysis on the impact of external factors such as weather and previous game outcomes on match predictions. Their study highlighted how contextual factors play a crucial role in deciding the results of matches [6].

Long Short-Term Memory (LSTM) networks were used by Sundar and associates (2021) to analyze sequential player performance data and forecast match results. Their method showed how well temporal models could capture changing game situations [7]. Additionally, convolutional neural networks (CNNs) have been explored for extracting spatial and temporal features from match highlights, demonstrating potential for enhanced player performance predictions.

Reinforcement learning was investigated by Gupta et al. (2022) to maximize tactics such as field placements and batting order. This approach has potential for real-time decision-making, even though it is mainly used in simulations [8]. Furthermore, multi-agent reinforcement learning frameworks are being developed to model team dynamics, enabling simulations that better reflect real-world scenarios.

According to Rajaraman et al. (2020), interpretability is crucial for prediction models. By applying SHAP (SHapley Additive exPlanations) values to cricket models, they were able to shed light on how factors such as player form, venue, and toss affect predictions [9]. Other explainability techniques, such as LIME (Local Interpretable Model-Agnostic Explanations), have also been employed to validate model predictions and provide insights into critical match-winning factors.

A number of other research indicate machine learning's expanding potential in cricket analytics:

Ahmed et al. (2020) created a hybrid model to forecast cricket results using ensemble techniques such as XGBoost. By include weather and pitch conditions in the feature set, they were able to increase the model's accuracy [10].
The application of ARNing in cricket analytics was investigated by Mukherjee and Mahapatra (2021), which enables models trained on one dataset to adapt to new datasets with less retraining. This method is especially helpful in situations with rare crickets [11].

By grouping player performance indicators using unsupervised techniques like k-means, Chawla et al. (2019) provided insights into player roles and team composition [12].

With an emphasis on factors like player injuries and unforeseen weather changes, Pandey et al. (2022) used Bayesian networks to model uncertainty in ch predictions [13].
Decision tree ensembles were used by Das and Roy (2020) to forecast IPL results, and they highlighted the value of feature engineering in raising prediction accuracy [14].

Verma et al. (2021) offered a fresh viewpoint on how fan attitude affects team morale and performance by incorporating sentiment analysis of social media data into match prediction [15].

Together, these studies underscore the possibilities and difficulties of utilizing machine learning for cricket analysis. They show that with machine learning models, prediction accuracy can be moderate, but by adding more variables and utilizing advanced techniques like feature selection, ensemble learning, and real-time data processing, prediction capabilities can substantially enhance.

IV. Dataset Analysis

Data has been collected from Kaggle [1]. The below table shows the data type and description of the Deliveries dataset.

Column Name	Data Type	Description
match_id	int	Unique ID for each match
inning	int	Inning number (1 or 2)
batting_team	Object	Name of the batting team
bowling_team	Object	Name of the bowling team
over	int	Over number (1-20 in T20 matches)
ball	int	Ball number in the over
batter	Object	Name of the batsman at the crease
bowler	Object	Name of the bowler
Non_stiker	Object	Name of the non-striking batsman
Batsman_runs	int	Runs scored by the batsman of the ball
Extra_runs	int	Extra runs scored on the ball (wide, no-balls etc)
Total_runs	int	Total runs scored from the ball, including extras
Extras_type	Object	Type of extra runs
Is_wicket	int	Indicates if a wicket was taken
Player_dismissed	Object	Name of the player dismissed
Dismissal_kind	Object	Type of dismissal (caught, bowled etc)
Fielder	Object	Name of the fielder involved in the dismissal (if applicable)

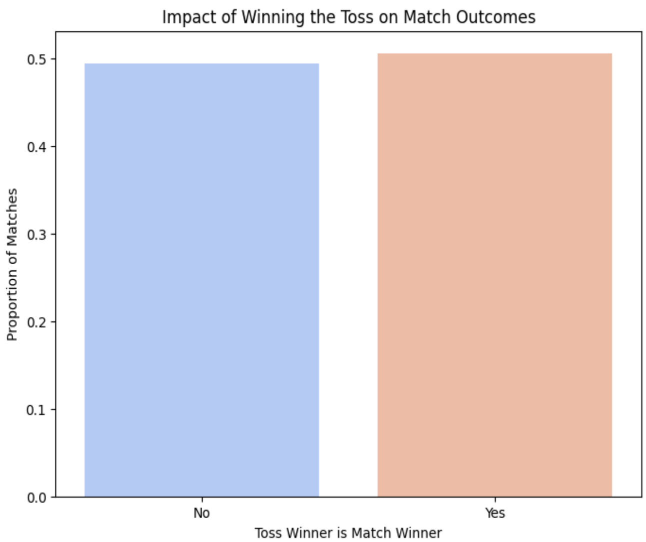
The below table shows the data type and description of the Matches dataset.
The graph 1 displays the percentage of cricket matches where the team that won the toss also won the match, compared to instances where they did not. The blue bar indicates the matches where the toss-winning team lost, and the salmon-colored bar shows where the toss-winning team won the match. If the bars are about the same height, it means that winning the toss generally doesn't greatly affect the final result of the match. However, if one bar is much taller than the other, it suggests that there's a significant link between winning the toss and winning the match.

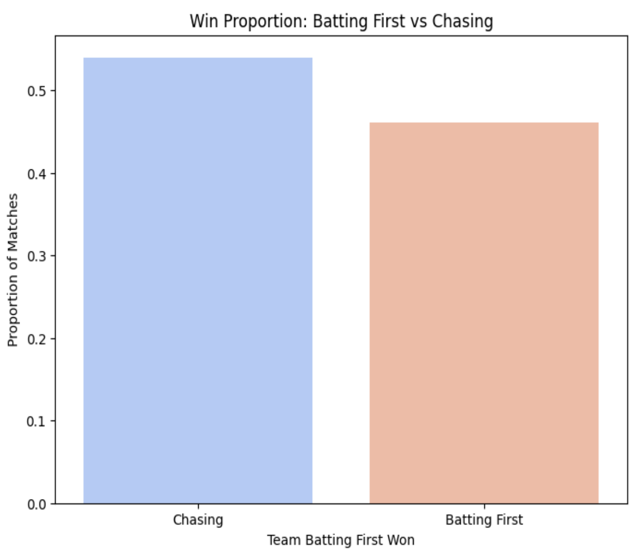
The graph 2 illustrates the win percentages of cricket teams that batted first against those that chased (batted second). The blue bar represents the proportion of matches won by teams that chased, whereas the salmon-colored bar represents the proportion of matches won by teams that batted first. Analyzing this graph helps in understanding which batting strategy might be more successful. A taller blue bar indicates that chasing is often more advantageous, while a taller salmon bar suggests that batting first can provide a competitive advantage.

Column Name	Data Type	Description
id	int	Unique ID for each match
city	Object	City where the match was played
date	Object	Date on which the match was played
Player_of_match	Object	Player who was named "Player of the Match"
Venue	Object	Venue where the match was held
Neutral_venue	int	Indicates if the match was at a neutral venue (1 for yes, 0 for no)
Team1	Object	One of the teams playing the match
Team2	Object	The other team playing the match
Toss_winner	Object	Team that won the toss
Toss_decision	Object	Decision taken by toss winner to 'bat' or 'field'
winner	Object	Team that won the match
result	Object	Total runs scored from the ball, including extras
Result_margin	float	Result of the match ('runs', 'wickets', 'tie', 'no result')
Target_runs	float	Target runs set by the first batting team
Target_overs	float	Overs in which the target runs were to be chased
Super_over	Object	Indicates if there was a super over ('Y' or 'N')
method	Object	Indicates if a special method was used in rainaffected games
Umpire1	Object	Primary umpire of the match
Umpire2	Object	Secondary umpire of the match

This graph 3 presents a bell-shaped distribution, showcasing the frequency of total runs scored in matches. The x-axis represents the total runs scored per match, while the y-axis indicates the frequency of these scores. The distribution peaks around the 250-300 runs mark, suggesting that most cricket matches end with scores in this range. This visualization helps in understanding the common scoring outcomes in matches, with the majority clustering around the central peak and fewer occurrences of extremely low or high scores.
The graph 4 illustrates the average number of runs scored in each over of a match. The x-axis labels the overs from 0 to 19, and the y-axis represents the average runs scored in these overs. The colors change progressively from deep purple to

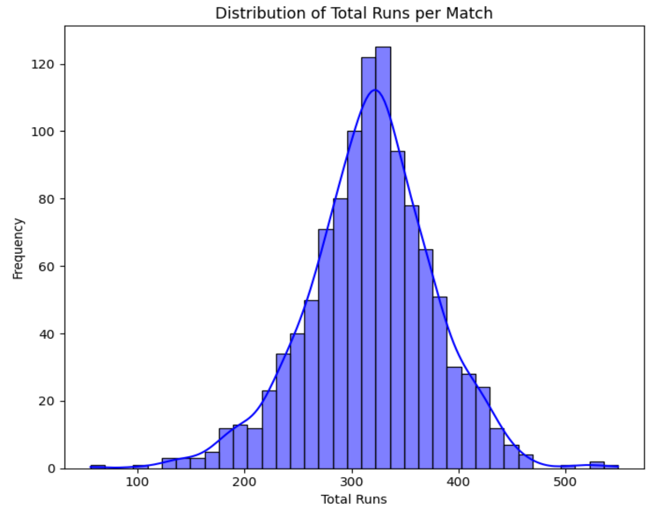
light green as the overs progress, indicating an increase in scoring rate. This pattern suggests that as the match progresses, particularly towards the final overs, the scoring rate tends to increase. This could be due to teams trying to maximize their score in the latter part of the innings, taking more risks as the game advances.

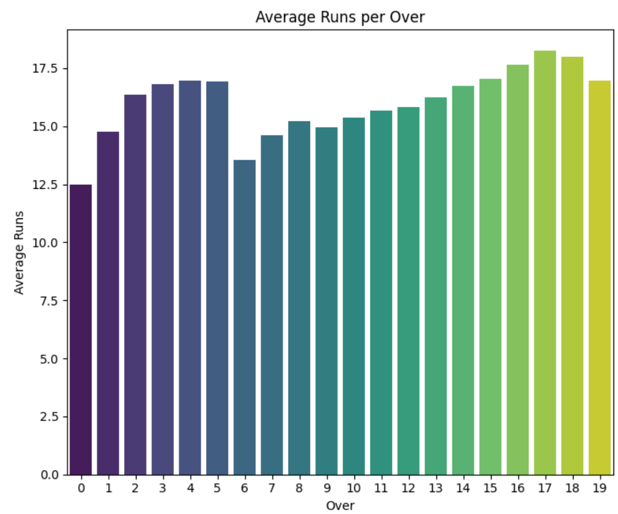




The histogram (graph 5) below shows the Powerplay Performance and Match Result based on the number of runs scored during the first 6 overs (Powerplay) and whether the team went on to win the match. There is a significant overlap between the red and teal bars, especially for teams scoring between 30-60 runs in the Powerplay. This indicates that while a strong Powerplay performance is important, it does not always guarantee a win.

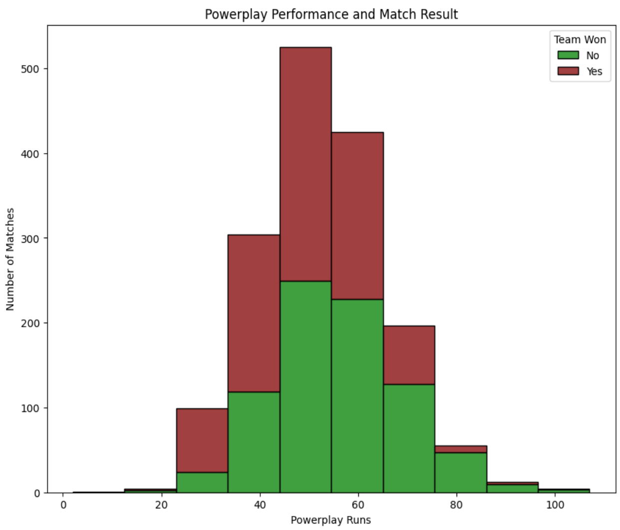
Teams that scored higher than 60 runs in the Powerplay have a greater tendency to win, as evidenced by the predominance of teal bars in that range. Teams scoring below 30 runs in the Powerplay generally lost the match, as indicated by the dominance of red bars in that range.



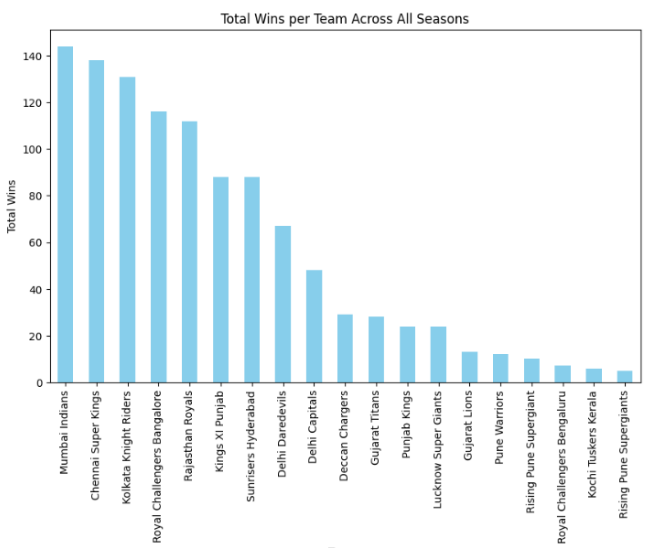


The bar graph (graph 6) displays notable performance differences by showing the total number of victories attained by different teams during every season of a cricket league. The teams with the most wins—the Mumbai Indians, Chennai Super Kings, and Kolkata Knight Riders—show constant supremacy, followed by the somewhat successful Royal Challengers Bangalore and Rajasthan Royals. While newer or short-lived teams like Gujarat Titans, Lucknow Super Giants, and Kochi Tuskers Kerala have relatively fewer wins, which reflects their limited participation or shorter franchise history, teams like Sunrisers Hyderabad, Delhi Capitals, and Deccan Chargers perform at a mid-level. This distribution draws attention to the disparities in the league's teams' performance consistency and competitive dynamics.

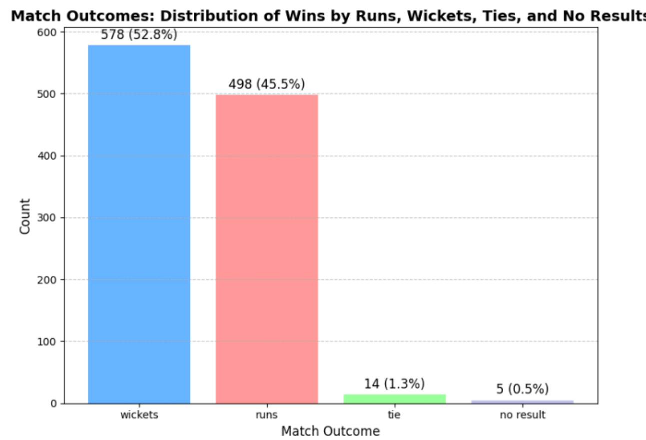
The bar chart (graph 7) depicts the distribution of cricket league match results is shown in the bar chart, which is divided into categories such as wins by wickets, runs, ties, and no results. The most common ways to win are by taking

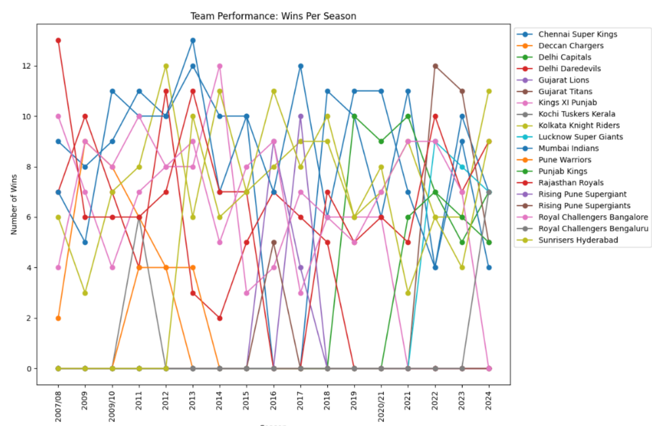


wickets, which account for 578 matches (52.8%), and by scoring runs, which account for 498 matches (45.5%). Only 14 matches (1.3%) conclude in a tie, while there are even fewer matches with no outcome (5 matches, 0.5%). Reflecting the tough and decisive nature of the league's games, the graphic highlights the balance between the two leading winning strategies while highlighting the rarity of draws and games with no outcome.



The Line chart (graph 7) performance of different cricket league teams is depicted in the line chart, which shows how many victories each team has throughout each season from 2008 to 2024. Season-long dominance is demonstrated by teams like the Chennai Super Kings and Mumbai Indians, who consistently perform well and reach significant win totals. Teams like the Gujarat Titans and the Lucknow Super Giants, on the other hand, have only performed in recent seasons, which is indicative of their more recent involvement. Teams with shorter franchise histories, like the Deccan Chargers and Kochi Tuskers Kerala, exhibit activity for brief periods of time. The league's dynamic and competitive nature is demonstrated by the chart's varying performances, which show certain teams remaining consistent while others display unpredictability.





V. Proposed Approach

Research Question 1: Does Winning the Toss Affect Match Outcome?

To find out if winning the toss has any impact on the match result, three models were tested: Naive Bayes, Random Forest, and Logistic Regression. A new binary target variable win_toss_win_match was created to indicate if the team that won the toss also won the match. If the toss winner is the same as the match winner, the value is 1 (Yes), otherwise, it is 0 (No).

For the Naive Bayes model, the toss winner was used to predict whether the toss-winning team would also win the match. The data was split into 70% for training and 30% for testing. The model's results were evaluated using a confusion matrix.

In the Random Forest model, additional details such as toss winner, match location, and team information were added as features. The Random Forest model used SMOTE to address the imbalance in the classes. After training, the model's performance was checked based on accuracy, precision, recall, and F1 scores.

For the Logistic Regression model, only the toss winner feature was used, converted into a binary format. The model was trained on 70% of the data and tested on 30%, aiming to predict if the toss-winning team would also win the match. Performance was measured using standard classification metrics.

Research Question 2: Role of Powerplay Performance in Determining Match Outcome

The second question aimed to understand if a team's performance in the powerplay runs scored and wickets lost in the first six overs affects the match result. Each model used similar features for this question.

In the Naive Bayes model, powerplay runs and wickets were the main features for predicting match outcomes as win or loss. The model was trained and tested on separate sets, and its performance was measured using a confusion matrix.

For the Random Forest model, powerplay runs and wickets were again used as predictors. SMOTE was applied to balance the data, and the model was trained to see if powerplay performance had any noticeable impact on the final match outcome.

In the Logistic Regression model, powerplay runs and wickets were also used as key features. These statistics were aggregated at the match level, and Logistic Regression was applied to predict match outcomes. The model was trained on 70% of the data and tested on 30%, with accuracy, precision, and recall used to evaluate its performance.

Research Question 3: Influence of Individual Performance Metrics on Player of the Match

The third question aimed to predict the "Player of the Match" award based on individual player metrics. Strike rate and economy rate were used to see if they could predict which player would win the award.

For the Naive Bayes model, strike rate and economy rate were the features used to predict if a player would receive the "Player of the Match" award. The model was evaluated with a confusion matrix to check its accuracy.

In the Random Forest model, additional player performance metrics like runs scored, strike rate, wickets taken, and economy rate were used. Due to the class imbalance (more non-winners than winners), SMOTE was applied to create a balanced dataset. The model was trained and its ability to distinguish between award winners and non-winners was evaluated.

Lastly, the Logistic Regression model used strike rate and economy rate to predict the likelihood of winning the "Player of the Match" award. The data was split into training and testing sets, and the model's performance was checked using accuracy, precision, recall, and F1 scores to see how well it could predict award winners compared to non-winners.

VI. PROPOSED EVALUATION METHOD

The proposed evaluation method for this study employs a combination of accuracy, precision, recall, and F1-score to assess the effectiveness of the models in addressing the research questions. These metrics provide a comprehensive view of prediction quality, with accuracy offering an overall success rate, precision highlighting correct predictions for each class, and recall measuring the ability to identify true instances. F1-score balances precision and recall, making it particularly useful for datasets with imbalanced classes.

To optimize model performance, Grid Search with Cross-Validation was employed to identify the best hyperparameters for each model. Additionally, learning curves were generated to evaluate model performance across varying training set sizes, providing insights into how well the model generalizes to unseen data. These methods ensure robust evaluation and help refine the models to achieve the best results for the given research questions. These combined metrics allow for a comprehensive assessment of each model's strengths and limitations in relation to the research questions.

VII. RESULTS ANALYSIS

Research Question 1: Does Winning the Toss Affect Match Outcome?

A. NAÏVE BAYES MODEL

The Naïve Bayes model is used to investigates the relationship between winning the toss and match outcomes. The model achieved an accuracy of 52%, indicating moderate predictive performance. The precision for predicting a "Win" is 0.58, while its recall is relatively low at 0.22, suggesting that the model struggles to identify toss winners who also win the match. Conversely, the "Loss" class has a recall of 0.83, demonstrating the model's strength in identifying instances where toss winners lose. The overall F1-scores for both classes highlight the challenge of achieving balanced predictions, with a macro-average F1-score of 0.48. These results suggest that winning the toss has limited predictive power for determining match outcomes.

	Precision	Recall	F1-score	Support
Loss	0.51	0.83	0.63	107
Win	0.58	0.22	0.32	112
Accuracy			0.52	219
Macro Avg	0.54	0.53	0.48	219
Weighted Avg	0.54	0.52	0.47	219

B. RANDOM FOREST CLASSIFIER

The Random Forest classifier output shows that the model's precision for predicting losses (Class 0) is 0.47, and for wins (Class 1) it's 0.49, which indicates the model has slightly below-average accuracy for both outcomes. The recall is 0.42 for losses and 0.53 for wins, meaning the model has limited effectiveness in recognizing actual match results. The F1-Scores are 0.44 for losses and 0.51 for wins, which shows that the model doesn't perform very well overall.

With an accuracy of 48%, the model does only a little better than random guessing, which suggests it may not be learning strong patterns from the data features. The macro and weighted averages for precision, recall, and F1-Score are around 0.48, meaning the model performs similarly across both classes, but not at a high level.

	Precision	Recall	F1-score	Support
Loss	0.47	0.42	0.44	162
Win	0.49	0.53	0.51	167
Accuracy			0.48	329
Macro Avg	0.48	0.48	0.48	329
Weighted Avg	0.48	0.48	0.48	329

C. LOGISTIC REGRESSION

The logistic regression model achieved an accuracy of approximately 57%, which indicates that it is not highly effective in predicting whether the toss winner will win the match. The precision for class 0 (not winning the match) was 0.61, and recall was 0.29, meaning the model struggles with predicting the non-winner class. The precision for class 1 (winning the match) was 0.56, with a recall of 0.82, suggesting that the model is better at predicting when the toss winner wins the match, but it is still not highly precise.

Overall, all the three model treat both classes fairly but doesn't predict match outcomes very accurately, suggesting that adding more features, adjusting the model settings, or trying different models could improve accuracy.

	Precision	Recall	F1-score	Support
Loss	0.61	0.29	0.39	158
Win	0.56	0.82	0.67	171
Accuracy			0.57	329
Macro Avg	0.58	0.56	0.53	329
Weighted Avg	0.58	0.57	0.53	329

Research Question 2: Role of Powerplay Performance in Determining Match Outcome

A. NAÏVE BAYES MODEL

For the Naive Bayes model addressing Research Question 2 (Role of Powerplay Performance in Determining Match Outcome), the model achieved an accuracy of 63%, indicating moderate predictive capability. The "Win" class demonstrated a precision, recall, and F1-score of 0.67, reflecting the model's ability to effectively identify winning teams based on their Powerplay performance. The "Loss" class showed balanced metrics, with a precision, recall, and F1-score of 0.59. Both the macro-average and weighted average metrics were consistent at 0.63, reinforcing the model's overall balanced performance. These findings highlight that Powerplay performance (runs scored and wickets lost) plays a moderately significant role in predicting match outcomes when using the Naive Bayes model.

	Precision	Recall	F1-score	Support
Loss	0.59	0.59	0.59	98
Win	0.67	0.67	0.67	121
Accuracy			0.63	219
Macro Avg	0.63	0.63	0.63	219
Weighted Avg	0.63	0.63	0.63	219

B. RANDOM FOREST CLASSIFIER

The Random Forest classifier output shows that for predicting losses (Class 0), the model has a precision, recall, and F1-Score of 0.53, while for predicting wins (Class 1), these metrics are slightly higher, at 0.58, 0.59, and 0.59, respectively.

The overall accuracy of the model is 56%, indicating moderate performance in predicting match outcomes. Both the macro and weighted averages for precision, recall, and F1-Score are also 0.56, suggesting the model performs consistently across both classes but at a modest level of accuracy. This result implies the model has some ability to predict outcomes based on the features used but may still need improvement to enhance predictive accuracy.

	Precision	Recall	F1-score	Support
Loss	0.53	0.52	0.52	154
Win	0.58	0.59	0.59	175
Accuracy			0.56	329
Macro Avg	0.56	0.56	0.56	329
Weighted Avg	0.56	0.56	0.56	329

C. LOGISTIC REGRESSION

The logistic regression model achieved an accuracy of 63%. The precision and recall for the match outcome (class 1, win) were 0.65, suggesting that the model is fairly balanced between predicting wins and losses. The precision and recall for losses (class 0) were 0.61, indicating the model performs somewhat equally across both classes, but it could be further refined to improve prediction accuracy for both outcomes.

	Precision	Recall	F1-score	Support
Loss	0.61	0.61	0.61	312
Win	0.65	0.65	0.65	345
Accuracy			0.63	657
Macro Avg	0.63	0.63	0.63	657
Weighted Avg	0.63	0.63	0.63	657

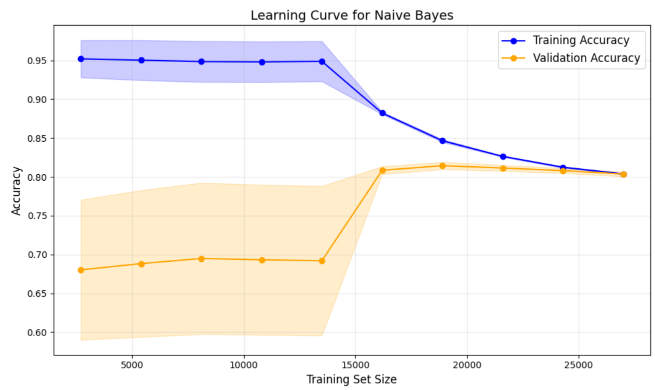
Research Question 3: Influence of Individual Performance Metrics on Player of the Match

A. NAÏVE BAYES MODEL

The Influence of Individual Performance Metrics on Player of the Match using the Naive Bayes model, the results show an accuracy of 83%, primarily due to strong performance in predicting non-award winners (precision: 0.98, recall: 0.84, F1-score: 0.91). However, the model struggled with award winners, achieving a precision of 0.18 and F1-score of 0.29, reflecting significant class imbalance. This highlights the need for improved data balancing or feature engineering to better predict award winners.

	Precision	Recall	F1-score	Support
Loss	0.98	0.84	0.91	5625
Win	0.18	0.72	0.29	274
Accuracy			0.83	5899
Macro Avg	0.58	0.78	0.60	5899
Weighted Avg	0.95	0.83	0.88	5899

The learning curve for Naive Bayes shows high training accuracy initially, indicating that the model fits the training data well. However, the gap between training and validation accuracy narrows as the training set size increases, suggesting improved generalization with more data. The model achieves stable performance as the validation accuracy plateaus near 0.8 with larger datasets.



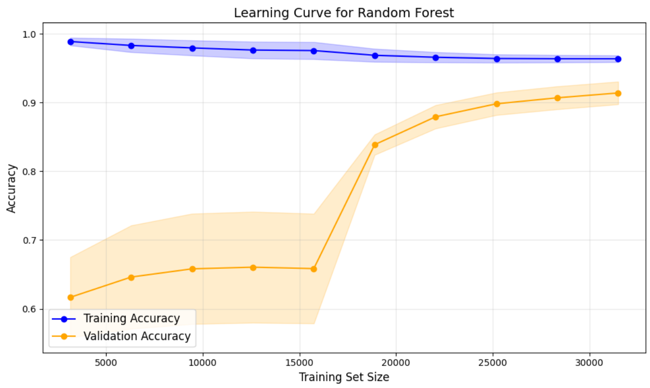
B. RANDOM FOREST MODEL

The Random Forest classifier output shows that for predicting "No Award," the model achieves high precision, recall, and F1-Score of 0.97, 0.94, and 0.96, respectively, indicating strong performance in identifying players who did not win the award. However, for predicting "Won Award," the model has much lower metrics, with a precision of 0.25, recall of 0.42, and F1-Score of 0.32, reflecting its difficulty in accurately identifying award winners.

The model's overall accuracy is 92%, largely driven by its ability to correctly classify the majority "No Award" class. The macro averages for precision, recall, and F1-Score are 0.61, 0.68, and 0.64, respectively, while the weighted averages are 0.94, 0.92, and 0.93. This shows that, while the model performs well overall, it is heavily biased toward the majority class and struggles with the minority class, indicating a need for further adjustments to better predict "Won Award" cases.

	Precision	Recall	F1-score	Support
No Award	0.97	0.94	0.96	8436
Won Award	0.25	0.42	0.32	412
Accuracy			0.92	8848
Macro Avg	0.61	0.68	0.64	8848
Weighted Avg	0.94	0.92	0.93	8848

The learning curve for Random Forest indicates that the model consistently achieves near-perfect training accuracy, suggesting a strong fit to the training data. Validation accuracy steadily improves with increasing training set size and stabilizes above 0.9, indicating excellent generalization. The small gap between training and validation accuracy at larger datasets reflects the model's robustness and minimal overfitting.



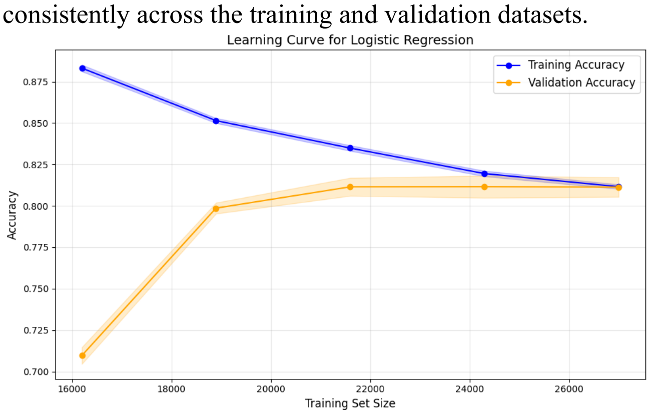
C. LOGISTIC REGRESSION

The logistic regression model achieved an accuracy of 87%, with a strong performance on the majority class (non-Player of the Match, class 0), showing high precision and recall for that class.

However, the model failed to predict the Player of the Match (class 1) with zero precision and recall, indicating a significant imbalance in the dataset that led the model to heavily favor the majority class. Further techniques like class balancing or resampling could be employed to improve performance for the minority class.

	Precision	Recall	F1-score	Support
Loss	0.87	1.00	0.93	14043
Win	0.00	0.00	0.00	2179
Accuracy			0.87	16222
Macro Avg	0.43	0.50	0.46	16222
Weighted Avg	0.75	0.87	0.80	16222

The learning curve shows a decreasing trend in training accuracy as the dataset size increases, which is expected as the model becomes more generalized. Validation accuracy steadily improves and stabilizes around 0.82, closely aligning with the training accuracy. This indicates that the model is well-generalized with minimal overfitting and performs consistently across the training and validation datasets.



VIII. CONCLUSION

The promise and difficulties of using machine learning to forecast IPL match results and player performances are illustrated by this effort. Although the models offer insightful information about the variables affecting game outcomes, they also draw attention to cricket's intrinsic volatility.

Key findings include:

- Impact of Toss: Models show relatively slight increases in predicted accuracy among models, suggesting that winning the toss has little effect on match results. This implies that other elements may be more important.
- Powerplay Performance: There is a moderate correlation between match results and powerplay performance (runs scored and wickets lost). The relative superior performance of the Naïve Bayes and logistic regression models suggests that this stage may be a useful predictive characteristic.
- Individual Metrics for Player Awards: Class disparities made it difficult to predict "Player of the Match" awards using individual indicators like strike rate and economy rate. Despite its potential, Random Forest had trouble with minority classes, highlighting the necessity for more thorough feature engineering.

IX. FUTURE WORK

For future work, this study could be expanded by incorporating more detailed and nuanced features to gain deeper insights into the factors influencing match outcomes and player performances. For instance, adding specific player statistics such as the frequency of boundaries, strike rates during different phases of the game (e.g., Powerplay, middle overs, and death overs), or individual performance under varying pitch and weather conditions could significantly enhance the predictive power of the models. Furthermore, analyzing specific game scenarios, such as player performance in high-pressure situations or during close finishes, could provide valuable contextual information that may refine the models' predictions.

Temporal analysis, which focuses on how players perform during different stages of a match or across multiple games in a season, could also offer a richer understanding of trends and patterns in performance. This type of analysis might reveal insights into consistency, adaptability, or momentum, which are critical in a dynamic sport like cricket.

Additionally, employing advanced machine learning approaches, such as ensemble methods (e.g., stacking or boosting techniques) or deep learning architectures, could help capture complex, non-linear relationships in the data. These methods may offer superior performance in handling large, high-dimensional datasets, enabling stronger and more accurate predictions across all research questions. Finally, integrating external data sources, such as live match updates, player fitness levels, or crowd behavior, could further broaden the scope of analysis and provide a more holistic view of the factors influencing cricket outcomes. This expanded approach could not only improve predictive accuracy but also offer actionable insights for teams, analysts, and stakeholders in cricket.

X. Project Timeline

Below tables shows our project timeline:

Date	Task
September 16	Finalize problem statement, research questions, and hypothesis. Complete preliminary literature review. Draft project proposal (3-4 pages).
September 17	Submit Project Proposal
September 24	Data collection, cleaning, and exploratory data analysis (EDA). Begin building initial models
October 7	**Submit project milestone 1**
October 8	Complete first round of analysis and gather preliminary results.
October 29	Start finalizing preliminary results. Prepare report for Milestone 2
November 11	**Submit Project Milestone 2**
November 19	Prepare the final report (IEEE format), project code, and presentation slides.
December 3	Review and submit the final project package (report, code, slides, website).
December 6	**Project submission deadline**

XI. References

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